Yash Joshi

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Education

- **Master of Science in Data Science** *Dec 2023 - Dec 2025*
  The University of Texas at Arlington

Experience

- **University of Texas at Arlington | Research Assistant | Dr. Prof. Erick C. Jones Jr** *Jan 2026 – Ongoing*
  - Prototyped a **Text-to-SQL** interface using **LLMs** enabling natural language querying of Industrial Lab Equipment structured databases, reducing analyst query-writing time by 40%
- **Larsen and Toubro Private LTD. | Data Engineer Intern** *Dec 2022 – May 2023*
  - Orchestrated automated data ingestion pipelines (Python, Pandas, SQLAlchemy, Airflow) connecting SAP ERP, SQL Server, and flat files, achieving 99.5% data integrity and reducing manual reporting prep by 25%.
  - Accelerated ETL pipeline performance by 79% (2 hours → 25 minutes) through parallel batch processing, SQL query optimization (indexing, join refactoring), and custom data quality scripts (schema drift detection, anomaly flagging), cutting production errors by 40%.
  - Implemented a document classification pipeline using Llama-2 and LangChain to automatically categorize 10,000+ unstructured procurement documents (invoices, POs, contracts), improving data retrieval speed by 60%.
- **Universal Recycle Solutions | AI & Machine Learning Engineer Intern | Startup** *February 2022 – July 2022*
  - Engineered automated ETL pipelines integrating SQL Server, Excel, and Power BI to track 6+ KPIs (waste volume, route efficiency, fuel usage), reducing costs by 27% and enabling real-time resource allocation across 12+ city zones.
  - Developed Python-based predictive analytics modules (Random Forest, Gradient Boosting) achieving 92% forecast accuracy for 5-8 day waste volume predictions, enabling proactive resource allocation and reducing manual planning overhead by 40%. Architected **R Shiny** dashboards leveraging dplyr and ggplot2 to visualize correlations between precipitation, seasonality, and waste weight, reducing data exploration time by 60% for city planners.

Personal Projects and Research 50+ projects

- **NegotiationArena - Reinforcement Learning Model - OpenEnv - unsloth.ai** *March 2026*
  - Built an Agentic AI SaaS negotiation environment using OpenEnv, Unsloth, and GRPO to train an LLM (Qwen2.5-7B-Instruct) that autonomously handles salary/equity/start-date discussions against 5 simulated hiring experts, with curriculum learning and Snorkel-weighted rewards; Dockerized runtime and Hugging Face Spaces deployment.
- **QuantsTrade AI** *Jan 2026*
  - **Architected** a hybrid 'GenAI Quant' ecosystem merging **XGBoost** forecasting with **Llama-3** sentiment analysis, implementing a **RAG** system to generate **embeddings** for 10k+ financial documents (10-Ks, news) for automated research.
  - **Synthesized** a real-time ML-LLM pipeline using **LangChain** and **Opus 4.5**, reducing data processing latency by **60%** via parallelized ETL jobs and achieving **85% backtested** directional accuracy on NASDAQ feeds. **Deployed** the architecture on **Neon Postgres (pgvector)**, optimizing **vector** retrieval to **<200ms** to enable instant contextualization of live market.
- **NASA Turbofan Engine Degradation Simulation (CMAPSS)**
  *Built a **predictive analytics** system to estimate **Remaining Useful Life** of an aircraft turbofan engine from real-world sensor data.*
  - **Engineered** an end-to-end predictive maintenance pipeline on the **NASA C-MAPSS dataset** (33K+ sensor cycles), reducing feature dimensionality (24→17) via variance thresholding to optimize training efficiency.
  - **Benchmarked** regression and classification models (XGBoost, SVR, Random Forest), achieving **sub-35 RMSE** for RUL prediction and **F1≥0.95** for failure classification. **Conducted** SHAP (SHapley Additive exPlanations) analysis to interpret complex sensor degradation patterns, directly enabling condition-based maintenance planning and operational risk mitigation.
- **Career Prediction System: Web application based on Multi class Classification Algorithm** *March 2025*
  Deployed ML algorithms to analyze user data and provide personalized career suggestions, addressing suitable job roles.
  - **Engineered** a multiclass career recommendation engine using **Random Forest** and **One-vs-Rest** strategies, employing **SMOTE/ADA-SYN** oversampling to resolve class imbalance and achieve **92% prediction accuracy**. **Deployed** a full-stack interface using **React.js** and **Neon Postgres** to serve model predictions in real-time, creating platform for career guidance.
  - **Published** research findings in **Springer's MIWAI 2023** conference proceedings and open-sourced the dataset on **Kaggle**, with **10K+ views** and **3.5K+ downloads**.

Technical Skills and Interests

Languages: Python, SQL (MySQL, PostgreSQL), Scala, Java, R, Bash, Shell Scripting, OpenEnv
AI & Machine Learning: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, BERT, Llama-3, RAG, LangChain.
Data Engineering & MLOps: Apache Airflow, Spark (PySpark), Kafka, Hadoop, Docker, Kubernetes, MLflow, CI/CD
Cloud & Databases: AWS (S3, SageMaker, BedRock Lambda), MongoDB, Firebase, Neon Postgres, SQLAlchemy, Hive
Visualization & Web: Tableau, Power BI, Matplotlib, Seaborn, Plotly, React.js, Node.js, Git, GitHub, RShiny

Leadership & Awards

Event	Role	Outcome
Meta x Celebray Valley Hackathon(March 2026)	Team Lead	**Top 6 Place**
Hack UTA (Apr 2025)	Team Lead	**1st Place**
State Level Paper Presentation (Sep 2019)	Presenter	**1st Place**
Smart India Hackathon (Dec 2022)	Team Leader	Finalist (**2nd Place**)